

JAVA ASSIGNMENT

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∴ Objective : To find the shortest path in an autogenerated matrix.

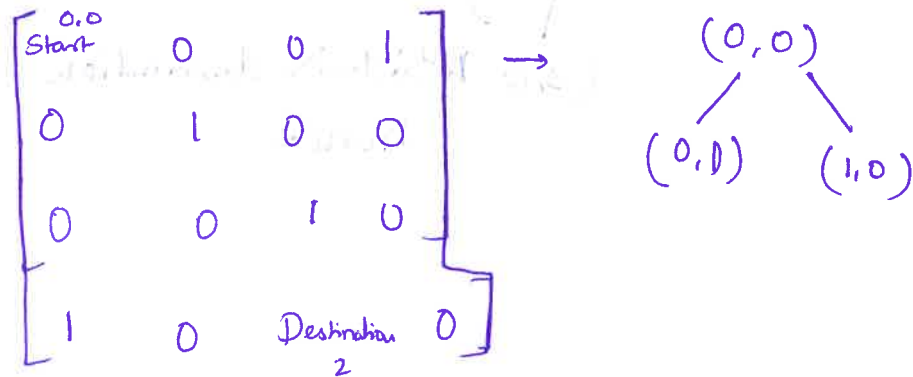
where: 0 is the free^{point}(path) which can be traversed
1 is the way which cannot be traversed.
2 is the destination.

Also mentioned that

- The user will provide matrix size
- User will provide the starting co-ordinate
- User will provide the destination coordinate:

Concept which I could use is BFS (Breadth First Search)

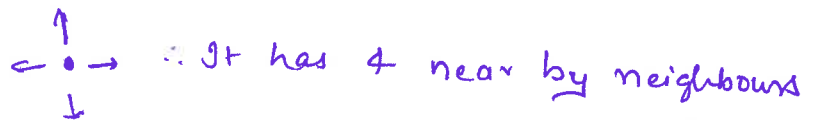
Suppose the generated grid or matrix is



Now if we convert this matrix into a tree then we will automatically get shortest path

Start (0,0)

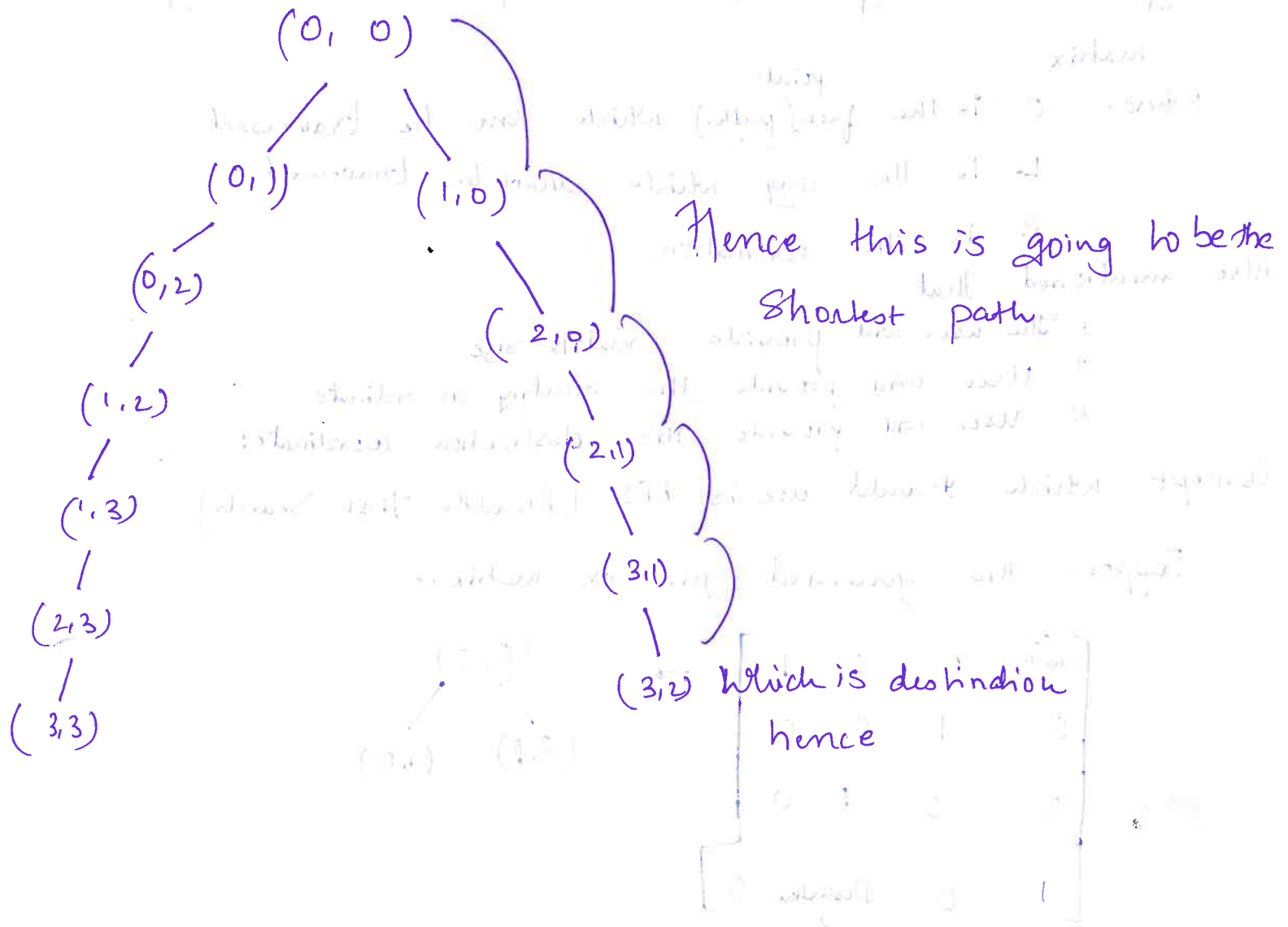
There are 4 ways a coordinate can be looked at



Now Start (0,0)

it can look in two ways (branching)

~~2~~. The tree diagram will look like



hence